Gustavo Rodriguez

Mechanical Engineering Student | **Expected Graduation: December 2028**
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Education

University of South Florida — **Tampa, FL**
Bachelor of Science in Mechanical Engineering, GPA: 3.33 — *Expected December 2028*
Recognition: Arthrex Scholar, USF Scholars Gold Recognition, Florida Academic Scholars.
Credentials: Lean Six Sigma Yellow Belt; CSWA - CAD Design; CSWA - Sustainability; ReCF Pre-Engineering & Robotics; CITI Research Integrity for Senior and Key Personnel; Altium Education PCB Basic Design Course Certificate.

Research & Engineering Leadership

IEEE EXO, University of South Florida — **Tampa, FL**
Project Lead / EXO Chair and President — *August 2025 – Present*

- Lead a multidisciplinary lower-limb exoskeleton program across **3 engineering subteams: Mechanical, Electrical, and Controls**; manage design reviews, technical deliverables, procurement, subsystem dependencies, testing, safety, and approximately **$5,000 in development budget**.
- Define requirements for a unilateral lower-limb **exoskeleton designed to provide up to 30% assistance at the hip and knee joints**; develop and test **MOV, EVT, and TPR control protocols** with **2 AK80-9 actuators**, IMUs, encoders, Jetson Orin Nano, and Teensy 4.1 controllers.
- Translate biomechanics and motion-analysis goals into **assistance timing, torque targets, stance/swing detection, and safety limits**; coordinate low-level **closed-loop torque/current control, CAN communication**, cross-team milestones, and technical documentation.

CARRT Lab, University of South Florida — **Tampa, FL**
REU Research Assistant — *August 2026 – Present*

- Develop and test an **IMU-based rugby lineout wearable** to study athlete motion and support **injury-prevention research**; integrate ESP32-based sensing with Vicon motion capture, validate sensor placement and enclosure fit, and support participant data collection and data-quality review.

Engineering Experience

Lonza — **Tampa, FL**
Reliability Engineering Intern — *August 2026 – December 2026*

- **Led the investigation** into recurring **Xcelodose 600S and microbalance failures** during a period when **75% of the equipment was down**; used inspection, alignment checks, load-cell/spring geometry, corner-load behavior, SOP review, and **6M, Gemba, and FMEA** to isolate failure pathways.
- Collaborated with **MSAT and Manufacturing** to translate investigation findings into corrective actions and develop a **retraining protocol** addressing identified failure modes, equipment handling, and recurrence prevention.
- Support Engineering, Reliability, Manufacturing, and Project Engineering with **equipment-risk reviews and validation/readiness activities**, technical-record review, process bottleneck investigations, and project-equipment decisions in a regulated manufacturing environment.

Facilities, Engineering & Maintenance Intern — *May 2026 – August 2026*

- Led a **Lean/5S A3** asset-identification improvement using standardized high-visibility labels and before/after time trials, reducing average equipment search time from roughly **9 minutes to 1.5 minutes (83%)**.
- Built digital maintenance and manufacturing workflows using **Power Apps, Power Automate, SharePoint/Lists, Forms, and Excel**, including a QR-linked tracker enforcing a **30-day equipment-cleaning interval** and structured operator/process records for traceability and shift continuity.
- Developed technician-facing maintenance training around Receive → Prepare → Execute → Test/Verify → Document → Close/Handoff; incorporated **GDP/ALCOA, escalation, work-order, and return-to-service expectations**.
- Supported five-year lifecycle/capital planning by cleaning and cross-referencing condition, risk, cost, ownership, and replacement data for **300+ equipment and infrastructure assets**.

Selected Technical Projects

Rugby Wearable Study Device | *SolidWorks, ESP32, IMU Sensors, 3D Printing*

- Redesigned the wearable from an initial approximately **80 × 60 mm concept** into a **62.17 × 58.00 × 16.50 mm battery-integrated modular assembly**, reducing footprint by approximately **25%** while adding a 40 × 10 × 4 mm battery, USB-C and switch access, ventilation, M2 fastening, reinforced supports, strap mounting, and serviceable modular interfaces for rugby lineout injury-prevention research.

Mime Robotic Hand | *Python, MediaPipe, ESP32, PCA9685, Servo Control*

- Integrated computer vision with an open-source **5-DOF tendon-driven hand**; developed **finger calibration, position mapping, smoothing/hysteresis, and reverse-servo logic**, plus a Rock-Paper-Scissors interaction mode.

S.M.U.R.F. - Shared Maintenance Urgent Request Form | *Power Apps, Power Automate, SharePoint/Lists*

- Built a **2-app** maintenance workflow with **5 concern-specific decision branches** to **decrease downtime and speed access to Engineering & Maintenance support** through automated urgency classification, alerts, assignment tracking, and searchable request records.

TRACE - Manufacturing Process Tracker | *Power Apps, SharePoint/Lists, Power Automate*

- Built a manufacturing run tracker around **3 live process states** (Active, Paused, Awaiting Handoff) to **increase manufacturing visibility, decrease downtime, and give operators quick access to support**; centralizes batch status, downtime capture, shift handoffs, notes, archive/void controls, and traceable activity history.

Technical Skills

Engineering / Electronics: SolidWorks 2025, Altium Designer, PCB design, schematic interpretation, assembly/drawing review, FDM printing, equipment validation, RCA/FMEA
Software / Controls: Python, MATLAB, Excel, Power Apps/Automate, SharePoint/Lists, Arduino/ESP32, Jetson/Teensy, sensors and actuators
Manufacturing / Quality: 5S/A3, GMP/GDP, SOPs, lifecycle/CapEx, visual management, technical documentation
Languages: English, Spanish